Experiment 2: Use of Input Statements and Operators

Aim : To understand input statements, arithmetic operations, and different operators in Python.

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1. Declare these variables (x, y and z) as integers. Assign a value of 9 to x. Assign a value of 7 to y, perform addition, multiplication, division and subtraction on these two variables and print out the result.

In [ ]:
```python
x = 9
y = 7

print("Addition:", x + y)
print("Multiplication:", x * y)
print("Division:", x / y)
print("Subtraction:", x - y)
```

Basic arithmetic operations are performed using operators like +, -, *, and /.

2. Write a Program where the radius is taken as input to compute the area of a circle.

In [ ]:
```python
radius = float(input("Enter radius: "))
area = 3.14 * radius * radius

print("Area of circle:", area)
```

The radius is taken as input and area is calculated using formula πr^2.

3. Write a Python program to solve (x+y)*(x+y)

In [ ]:
```python
x = int(input("Enter value of x: "))
y = int(input("Enter value of y: "))

result = (x + y) * (x + y)

print("Result:", result)
```

This program takes two inputs and calculates $(x + y)^2$ using multiplication.

4. Test data: x = 4, y = 3

5. Expected output: 49

In [ ]:
```python
x = 4
y = 3

result = (x + y) * (x + y)

print("Result:", result)
```

Here, fixed values are assigned to x and y as given in the test data.

The result of $(4 + 3)^2 = 49$, which matches the expected output.

6. Write a program to compute the length of the hypotenuse (c) of a right triangle using Pythagoras theorem.

In [ ]:
```python
a = float(input("Enter side a: "))
b = float(input("Enter side b: "))

c = (a**2 + b**2) ** 0.5
print("Hypotenuse:", c)
```

This program calculates the hypotenuse of a right triangle using Pythagoras theorem.

Formula: $c = \sqrt{(a^2 + b^2)}$

- ** is used for power.
- 0.5 power is used to calculate square root.

7. Write a program to find simple interests.

In [ ]:
```python
p = float(input("Enter principal: "))
r = float(input("Enter rate: "))
t = float(input("Enter time: "))

si = (p * r * t) / 100

print("Simple Interest:", si)
```

This program calculates simple interest using the formula:

SI = (P × R × T) / 100

Where: P = Principal, R = Rate, T = Time

8. Write a program to find area of triangle when length of sides are given.

In [ ]:
```python
a = float(input("Enter side a: "))
b = float(input("Enter side b: "))
c = float(input("Enter side c: "))

s = (a + b + c) / 2

area = (s * (s - a) * (s - b) * (s - c)) ** 0.5

print("Area of triangle:", area)
```

This program calculates the area of a triangle using Heron's formula.

Step 1: Calculate semi-perimeter: s = (a + b + c) / 2

Step 2: Area: $\sqrt{(s(s-a)(s-b)(s-c))}$

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9. Write a program to convert given seconds into hours, minutes and remaining seconds.

In [ ]:
```python
seconds = int(input("Enter seconds: "))

hours = seconds // 3600
minutes = (seconds % 3600) // 60
sec = seconds % 60

print("Hours:", hours)
print("Minutes:", minutes)
print("Seconds:", sec)
```

This program converts total seconds into hours, minutes, and seconds.

- // is floor division.
- % gives remainder.

1 hour = 3600 seconds 1 minute = 60 seconds

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10. Write a program to swap two numbers without taking additional variable.

In [ ]:
```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

a, b = b, a

print("After swapping:")
print("a =", a)
print("b =", b)
```

This program swaps two variables without using a third variable.

Python allows multiple assignment: a, b = b, a

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11. Write a program to find sum of first n natural numbers.

In [ ]:
```python
n = int(input("Enter n: "))

sum_n = n * (n + 1) // 2

print("Sum:", sum_n)
```

This program calculates the sum of first n natural numbers using formula:

Sum = n(n + 1) / 2

It avoids loops and is efficient.

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12. Write a program to print truth table for bitwise operators (&, |, and ^ operators)

In [ ]:
```python
print("A B | AND OR XOR")

for a in [0, 1]:
    for b in [0, 1]:
        print(a, b, "|", a & b, a | b, a ^ b)
```

This program prints the truth table for bitwise operators:

& → AND
| → OR
^ → XOR

It checks all combinations of 0 and 1.

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13. Write a program to find left shift and right shift values of a given number.

In [ ]:
```python
num = int(input("Enter number: "))

print("Left Shift (num << 1):", num << 1)
print("Right Shift (num >> 1):", num >> 1)
```

- Left shift (<<) multiplies the number by 2.
- Right shift (>>) divides the number by 2.

Example: 5 << 1 = 10
5 >> 1 = 2

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14. Using membership operator find whether a given number is in sequence (10,20,56,78,89)

In [ ]:
```python
nums = (10, 20, 56, 78, 89)

x = int(input("Enter number: "))

print("Is number present?", x in nums)
```

The 'in' operator checks whether a value exists in a sequence.

Returns: True → if present
False → if not present

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15. Using membership operator find whether a given character is in a string.

In [ ]:
```python
string = "Hello Python"

ch = input("Enter character: ")

print("Is character present?", ch in string)
```

This program checks whether a character exists in a string using 'in' operator.